1. The output of an AND gate with three inputs, A, B, and C, is HIGH when _____.
   A. A = 1, B = 1, C = 0
   B. A = 0, B = 0, C = 0
   C. A = 1, B = 1, C = 1
   D. A = 1, B = 0, C = 1
   Answer: Option C

2. If a 3-input NOR gate has eight input possibilities, how many of those possibilities will result in a HIGH output?
   A. 1 B. 2
   C. 7 D. 8
   Answer: Option A

3. If a signal passing through a gate is inhibited by sending a LOW into one of the inputs, and the output is HIGH, the gate is a(n):
   A. AND B. NAND
   C. NOR D. OR
   Answer: Option B

4. A device used to display one or more digital signals so that they can be compared to expected timing diagrams for the signals is a:
   A. DMM
   B. spectrum analyzer
   C. logic analyzer
   D. frequency counter
   Answer: Option C

5. When used with an IC, what does the term "QUAD" indicate?
   A. 2 circuits
   B. 4 circuits
   C. 6 circuits
   D. 8 circuits
   Answer: Option B

6. The output of an OR gate with three inputs, A, B, and C, is LOW when _____.
   A. A = 0, B = 0, C = 0
   B. A = 0, B = 0, C = 1
   C. A = 0, B = 1, C = 1
   D. all of the above
   Answer: Option A

7. Which of the following logical operations is represented by the + sign in Boolean algebra?
   A. inversion B. AND
   C. OR D. complementation
   Answer: Option C

8. Output will be a LOW for any case when one or more inputs are zero for a(n):
   A. OR gate
   B. NOT gate

C. AND gate
D. NOR gate
Answer: Option C

9. How many pins does the 4049 IC have?
A. 14 B. 16
C. 18 D. 20
Answer: Option B

10. Which of the following choices meets the minimum requirement needed to create specialized waveforms that are used in digital control and sequencing circuits?
A. basic gates, a clock oscillator, and a repetitive waveform generator
B. basic gates, a clock oscillator, and a Johnson shift counter
C. basic gates, a clock oscillator, and a DeMorgan pulse generator
D. basic gates, a clock oscillator, a repetitive waveform generator, and a Johnson shift counter
Answer: Option A

11. TTL operates from a _____.
A. 9-volt supply
B. 3-volt supply
C. 12-volt supply
D. 5-volt supply
Answer: Option D

12. The output of a NOR gate is HIGH if _____.
A. all inputs are HIGH
B. any input is HIGH
C. any input is LOW
D. all inputs are LOW
Answer: Option D

13. The switching speed of CMOS is now _____.
A. competitive with TTL
B. three times that of TTL
C. slower than TTL
D. twice that of TTL
Answer: Option A

14. The format used to present the logic output for the various combinations of logic inputs to a gate is called a(n):
A. Boolean constant
B. Boolean variable
C. truth table
D. input logic function
Answer: Option C

15. The power dissipation, PD, of a logic gate is the product of the _____.
A. dc supply voltage and the peak current

B. dc supply voltage and the average supply current
C. ac supply voltage and the peak current
D. ac supply voltage and the average supply current
Answer: Option B

16. A logic probe is again applied to the pins of a 7421 IC with the following results. Is there a problem with the circuit and if so, what is the problem?

PIN	INDICATOR	PIN	INDICATOR
1	ON	14	ON
2	PULSING	13	ON
3	DIM	12	ON
4	ON	11	DIM
5	ON	10	OFF
6	PULSING	9	PULSING
7	OFF	8	OFF

A. Pin 6 should be ON.
B. Pin 8 should be ON.
C. Pin 8 should be pulsing.
D. no problem
Answer: Option D

17. If a 3-input AND gate has eight input possibilities, how many of those possibilities will result in a HIGH output?
A. 1 B. 2
C. 7 D. 8
Answer: Option A

18. The Boolean expression for a 3-input AND gate is _____.
A. X = AB
B. X = ABC
C. X = A + B + C
D. X = AB + C
Answer: Option B

19. A CMOS IC operating from a 3-volt supply will consume _____.
A. less power than a TTL IC
B. more power than a TTL IC
C. the same power as a TTL IC
D. no power at all
Answer: Option A

20. What does the small bubble on the output of the NAND gate logic symbol mean?
A. open collector output
B. tristate
C. The output is inverted.
D. none of the above

Answer: Option C

21. What are the pin numbers of the outputs of the gates in a 7432 IC?
A. 3, 6, 10, and 13
B. 1, 4, 10, and 13
C. 3, 6, 8, and 11
D. 1, 4, 8, and 11
Answer: Option C

22. The output of a NOT gate is HIGH when _____.
A. the input is LOW
B. the input is HIGH
C. power is applied to the gate's IC
D. power is removed from the gate's IC
Answer: Option A

23. If the input to a NOT gate is A and the output is X, then _____.
A. X = A
B.
C. X = 0
D. none of the above
Answer: Option B

24. A logic probe is used to test the pins of a 7411 IC with the following results. Is there a problem with the chip and if so, what is the problem?

PIN	INDICATOR	PIN	INDICATOR
1	ON	14	ON
2	PULSING	13	ON
3	PULSING	12	PULSING
4	ON	11	ON
5	ON	10	OFF
6	OFF	9	ON
7	OFF	8	OFF

A. Pin 6 should be ON.
B. Pin 6 should be pulsing.
C. Pin 8 should be ON.
D. no problem
Answer: Option B

25. How many inputs of a four-input AND gate must be HIGH in order for the output of the logic gate to go HIGH?
A. any one of the inputs
B. any two of the inputs
C. any three of the inputs
D. all four inputs
Answer: Option D

26. If the output of a three-input AND gate must be a logic LOW, what must the condition of the inputs be?
A. All inputs must be LOW.
B. All inputs must be HIGH.
C. At least one input must be LOW.
D. At least one input must be HIGH.
Answer: Option C

27. Logically, the output of a NOR gate would have the same Boolean expression as a(n):
A. NAND gate immediately followed by an inverter
B. OR gate immediately followed by an inverter
C. AND gate immediately followed by an inverter
D. NOR gate immediately followed by an inverter
Answer: Option B

28. A logic probe is placed on the output of a gate and the display indicator is dim. A pulser is used on each of the input terminals, but the output indication does not change. What is wrong?
A. The dim indication on the logic probe indicates that the supply voltage is probably low.
B. The output of the gate appears to be open.
C. The dim indication is the result of a bad ground connection on the logic probe.
D. The gate is a tristate device.
Answer: Option B

29. What is the Boolean expression for a three-input AND gate?
A. $X = A + B + C$
B. $X = A \cdot B \cdot C$
C. $A – B – C$
D. A $ B $ C
Answer: Option B

30. Which of the following gates has the exact inverse output of the OR gate for all possible input combinations?
A. NOR B. NOT
C. NAND D. AND
Answer: Option A

31. What is the difference between a 7400 and a 7411 IC?
A. 7400 has two four-input NAND gates; 7411 has three three-input AND gates
B. 7400 has four two-input NAND gates; 7411 has three three-input AND gates
C. 7400 has two four-input AND gates; 7411 has three three-input NAND gates
D. 7400 has four two-input AND gates; 7411 has three three-input NAND gates
Answer: Option B

32. Write the Boolean expression for an inverter logic gate with input C and output Y.
A. $Y = C$
B. $Y = \overline{C}$

Answer: Option B

33. The output of an exclusive-OR gate is HIGH if _____.
A. all inputs are LOW
B. all inputs are HIGH
C. the inputs are unequal
D. none of the above
Answer: Option C

34. A clock signal with a period of 1 s is applied to the input of an enable gate. The output must contain six pulses. How long must the enable pulse be active?
A. Enable must be active for 0 s.
B. Enable must be active for 3 s.
C. Enable must be active for 6 s.
D. Enable must be active for 12 s.
Answer: Option C

35. The AND function can be used to _____ and the OR function can be used to _____ .
A. enable, disable
B. disable, enable
C. enable or disable, enable or disable
D. detect, invert
Answer: Option C

36. One advantage TTL has over CMOS is that TTL is _____.
A. less expensive
B. not sensitive to electrostatic discharge
C. faster
D. more widely available
Answer: Option B

37. A 2-input NOR gate is equivalent to a _____.
A. negative-OR gate
B. negative-AND gate
C. negative-NAND gate
D. none of the above
Answer: Option B

38. If a 3-input OR gate has eight input possibilities, how many of those possibilities will result in a HIGH output?
A. 1 B. 2
C. 7 D. 8
Answer: Option C

39. Fan-out is specified in terms of _____.
A. voltage
B. current
C. wattage
D. unit loads
Answer: Option D

40. How many input combinations would a truth table have for a six-input AND gate?

A. 32 B. 48
C. 64 D. 128
Answer: Option C

41. What is the circuit number of the IC that contains four two-input AND gates in standard TTL?
A. 7402 B. 7404
C. 7408 D. 7432
Answer: Option C

42. The terms "low speed" and "high speed," applied to logic circuits, refer to the _____.
A. rise time
B. fall time
C. propagation delay time
D. clock speed
Answer: Option C

43. The NOR logic gate is the same as the operation of the _____ gate with an inverter connected to the output.
A. OR
B. AND
C. NAND
D. none of the above
Answer: Option A

44. The logic expression for a NOR gate is _____.
A. $X = \overline{A} + B$
B. $X = A + \overline{B}$
C. $X = A + B$
D. $X = \overline{A + B}$
Answer: Option D

45. With regard to an AND gate, which statement is true?
A. An AND gate has two inputs and one output.
B. An AND gate has two or more inputs and two outputs.
C. If one input to a 2-input AND gate is HIGH, the output reflects the other input.
D. A 2-input AND gate has eight input possibilities.
Answer: Option C

46. The term "hex inverter" refers to:
A. an inverter that has six inputs
B. six inverters in a single package
C. a six-input symbolic logic device
D. an inverter that has a history of failure
Answer: Option B

47. How many inputs are on the logic gates of a 74HC21 IC?
A. 1 B. 2

C. 3 D. 4
Answer: Option D

48. The basic logic gate whose output is the complement of the input is the:
A. OR gate
B. AND gate
C. inverter
D. comparator
Answer: Option C

49. When reading a Boolean expression, what does the word "NOT" indicate?
A. the same as
B. inversion
C. high
D. low
Answer: Option B

50. The output of an exclusive-NOR gate is HIGH if _____.
A. the inputs are equal
B. one input is HIGH, and the other input is LOW
C. the inputs are unequal
D. none of the above
Answer: Option A

51. How many AND gates are found in a 7411 IC?
A. 1 B. 2
C. 3 D. 4
Answer: Option C

52. Which of the following equations would accurately describe a four-input OR gate when A = 1, B = 1, C = 0, and D = 0?
A. 1 + 1 + 0 + 0 = 01
B. 1 + 1 + 0 + 0 = 1
C. 1 + 1 + 0 + 0 = 0
D. 1 + 1 + 0 + 0 = 00
Answer: Option B

53. What is the name of a digital circuit that produces several repetitive digital waveforms?
A. an inverter
B. an OR gate
C. a Johnson shift counter
D. an AND gate
Answer: Option C

54. The basic types of programmable arrays are made up of _____.
A. AND gates
B. OR gates
C. NAND and NOR gates
D. AND gates and OR gates
Answer: Option D

55. The logic gate that will have HIGH or "1" at its output when any one (or more) of its inputs is HIGH is a(n):
A. OR gate
B. AND gate
C. NOR gate
D. NOT operation
Answer: Option A

56. CMOS IC packages are available in _____.
A. DIP configuration
B. SOIC configuration
C. DIP and SOIC configurations
D. neither DIP nor SOIC configurations
Answer: Option C

57. Which of the following is not a basic Boolean operation?
A. OR B. NOT
C. AND D. FOR
Answer: Option D

58. Which of the following gates is described by the expression ?
A. OR B. AND
C. NOR D. NAND
Answer: Option D

59. What is the Boolean expression for a four-input OR gate?
A. Y = A + B + C + D
B. Y = A· B · C · D
C. Y = A – B – C – D
D. Y = A $ B $ C $ D
Answer: Option A

60. How many truth table entries are necessary for a four-input circuit?
A. 4 B. 8
C. 12 D. 16
Answer: Option D

61. How many entries would a truth table for a four-input NAND gate have?
A. 2 B. 8
C. 16 D. 32
Answer: Option C

62. The Boolean expression for a 3-input OR gate is _____.
A. X = A + B
B. X = A + B + C
C. X = ABC
D. X = A + BC
Answer: Option B

63. From the truth table for a three-input NOR gate, what is the only condition of inputs A, B, and C that will make the output X high?

A. A = 1, B = 1, C = 1
B. A = 1, B = 0, C = 0
C. A = 0, B = 0, C = 1
D. A = 0, B = 0, C = 0

64. The logic gate that will have a LOW output when any one of its inputs is HIGH is the:

A. NAND gate
B. AND gate
C. NOR gate
D. OR gate

Answer: Option C

65. The output of a NAND gate is LOW if _____.

A. all inputs are LOW
B. all inputs are HIGH
C. any input is LOW
D. any input is HIGH

Answer: Option B